Music Application Development using Sentiment Analysis

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Introduction

- In today's digital age, music plays a significant role in people's lives, serving as a source of emotional expression, relaxation, and motivation.
- However, discovering music that resonates with one's current emotional state can sometimes be challenging.
- To address this issue, we are developing an Emotion-Based Music Recommendation Application.
- This application aims to provide users with personalized music suggestions that align with their current emotions, enhancing their music listening experience.

Purpose and Significance:

- The purpose of this application is to bridge the gap between users' emotional states and their music choices.
- By leveraging advanced natural language processing techniques and the Spotify API, the application will analyze user-input emotions and recommend songs that match those emotions.

- This is particularly valuable because:

Enhanced Music Exploration: Users will be introduced to new songs that match their current mood, leading to a more diverse and enjoyable music library.

Discovery of New Artists: Users can discover lesser-known artists whose music resonates with their emotions, fostering a sense of connection to the music community.

Emotional Well-being: Music has the power to influence emotions. The application can contribute to users' emotional well-being by suggesting songs that align with their desired emotional state, whether it's relaxation, motivation, or joy.

Time-Saving: Instead of spending time searching for appropriate music, users can rely on the application to quickly provide relevant recommendations.

Real-Life Problems Solved

Music Overwhelm

- The abundance of music choices can overwhelm users. The application streamlines music discovery by suggesting songs that match their emotions, saving time and reducing decision fatigue.

Mood Enhancement

- Music has a profound impact on mood. By offering music that resonates with users' emotions, the application contributes to their emotional well-being and mental health.

Exploration and Diversity

- Users often stick to familiar songs. The application exposes users to a variety of music genres and artists, encouraging exploration and diversification of musical tastes.

Connection to Music

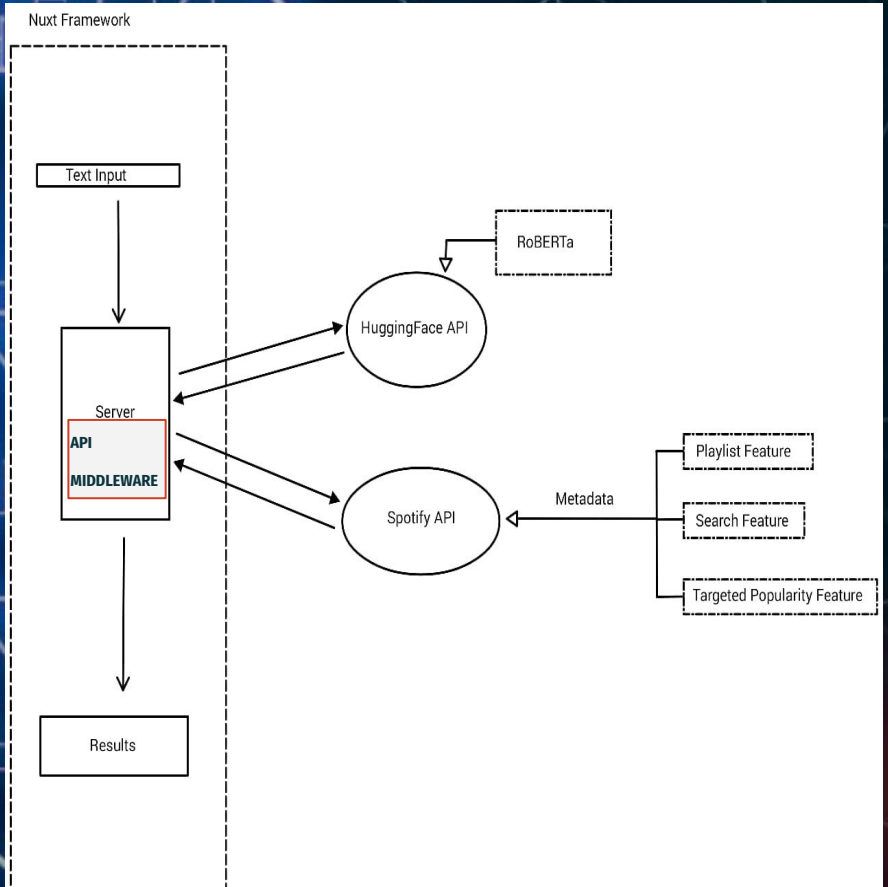
- Discovering music that aligns with one's feelings creates a stronger emotional connection between the listener and the music, leading to a more immersive music experience.

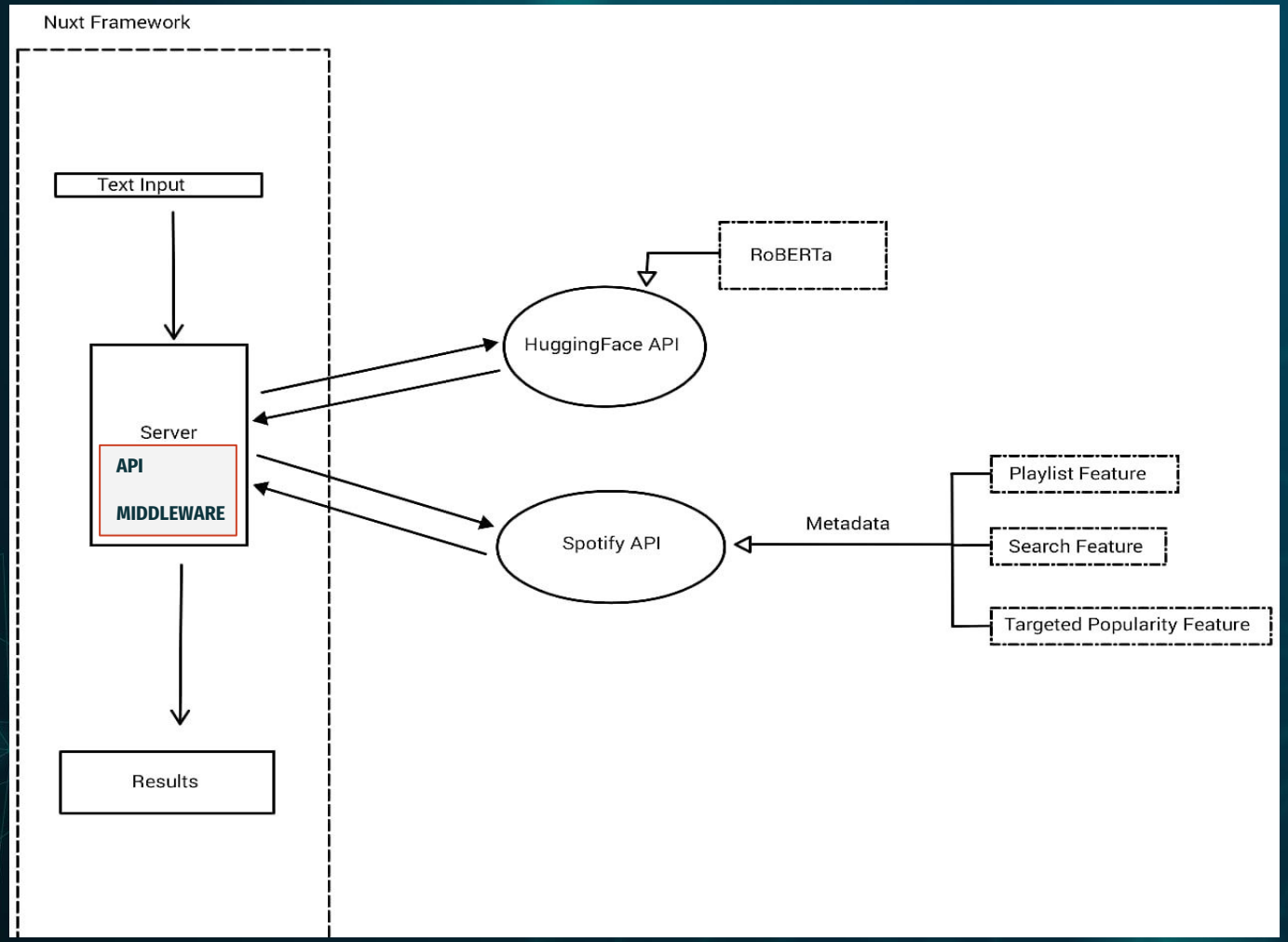
Application Workflow

1. Natural Language Processing
   - Huggingface
   - roBERTa

2. Spotify API Integration
   - List of usable Metadata
   - Reason to use spotify API

3. User Interface
   - Nuxt.js Framework



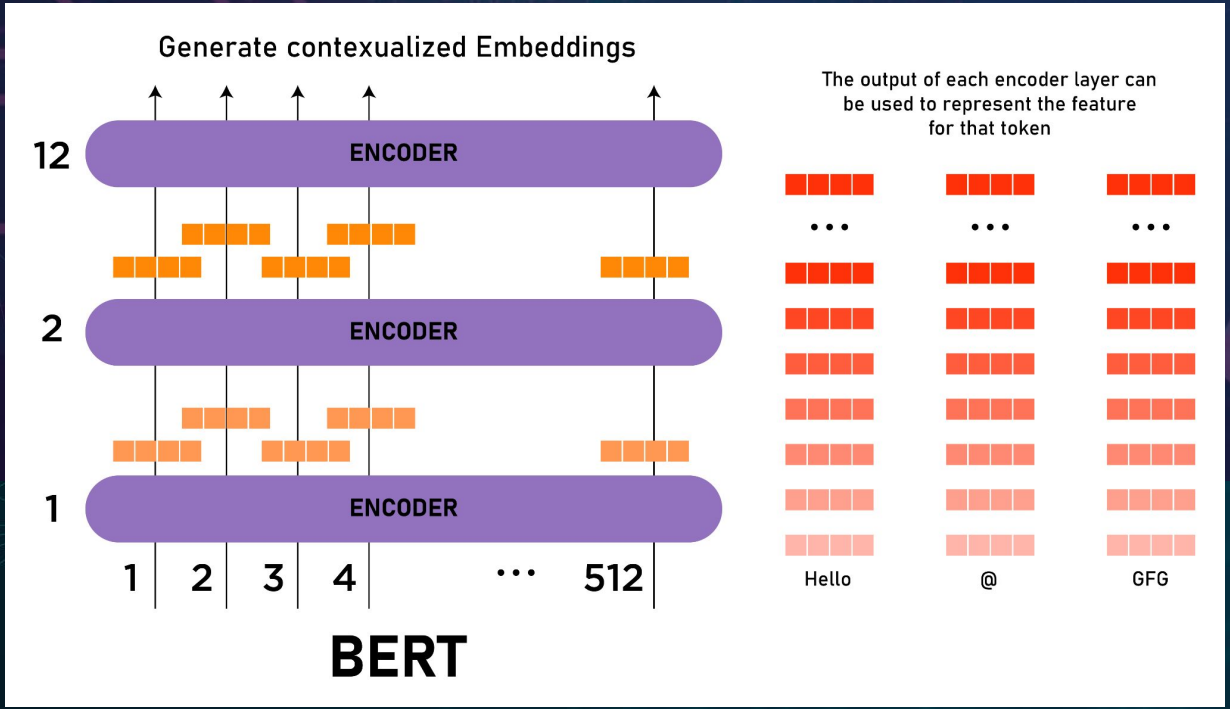

Nuxt Framework
Text Input
Server
API
MIDDLEWARE
Results
RoBERTa
HuggingFace API
Spotify API
Metadata
Playlist Feature
Search Feature
Targeted Popularity Feature

Hugging Face API Integration

Easily integrate NLP, audio and computer vision models deployed for inference via simple API calls. Harness the power of machine learning while staying out of MLOps!

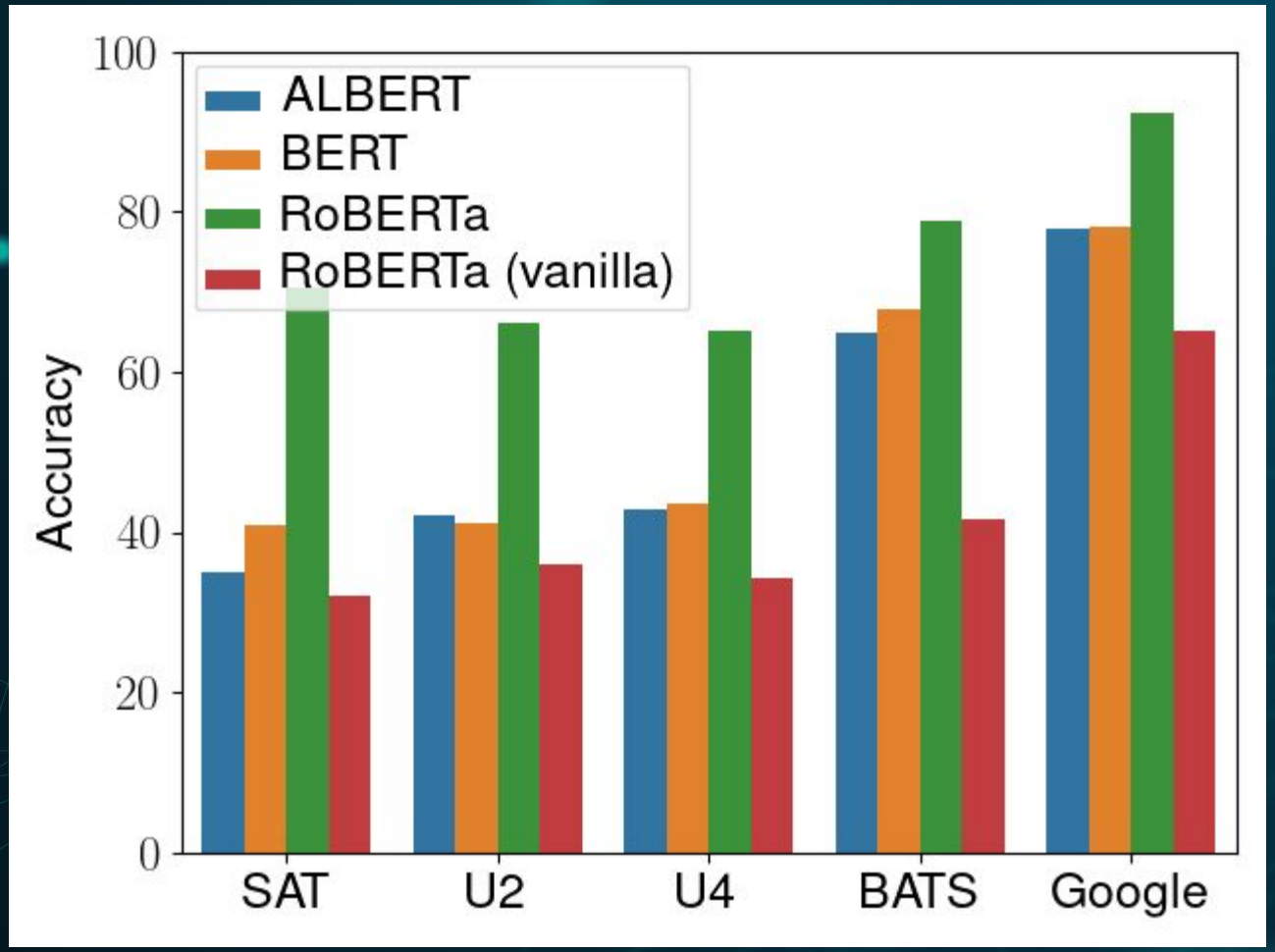
HUGGING FACE

BERT-base-model



Twitter-roBERTa-base-sentiment

The roBERTa-base model trained on ~58 million tweets and fine tuned for sentiment analysis with the tweetEval benchmark.


ALBERT
BERT
RoBERTa
RoBERTa (vanilla)
Accuracy
100
80
60
40
20
0
SAT
U2
U4
BATS
Google

Spotify for Developers

Spotify API Integration

- The application uses the Spotify API. The Spotify API provides a wide range of metadata (meta) that you can access to retrieve information about music, playlists, artists, albums, and more.
- It leverages the API's search and recommendation endpoints.

Here's a list of some of the metadata that the Spotify API offers:

1. Tracks:

- Track name
- Track ID
- Album name and ID
- Artist names and IDs
- Duration
- Popularity
- Preview URL
- Release date
- Explicit content flag
- External URLs (e.g., Spotify, Apple Music)
- Audio features (danceability, energy, etc.)

1. Albums:

- Album name
- Album ID
- Artist names and IDs
- Release date
- Total tracks
- Album cover images
- External URLs

1. Artists:

- Artist name
- Artist ID
- Popularity
- Genres
- Followers count
- Artist images
- External URLs

1. Playlists:

- Playlist name
- Playlist ID
- Owner information (user or artist)
- Total tracks
- Playlist cover images
- Public or private status
- External URLs
- Collaborative status

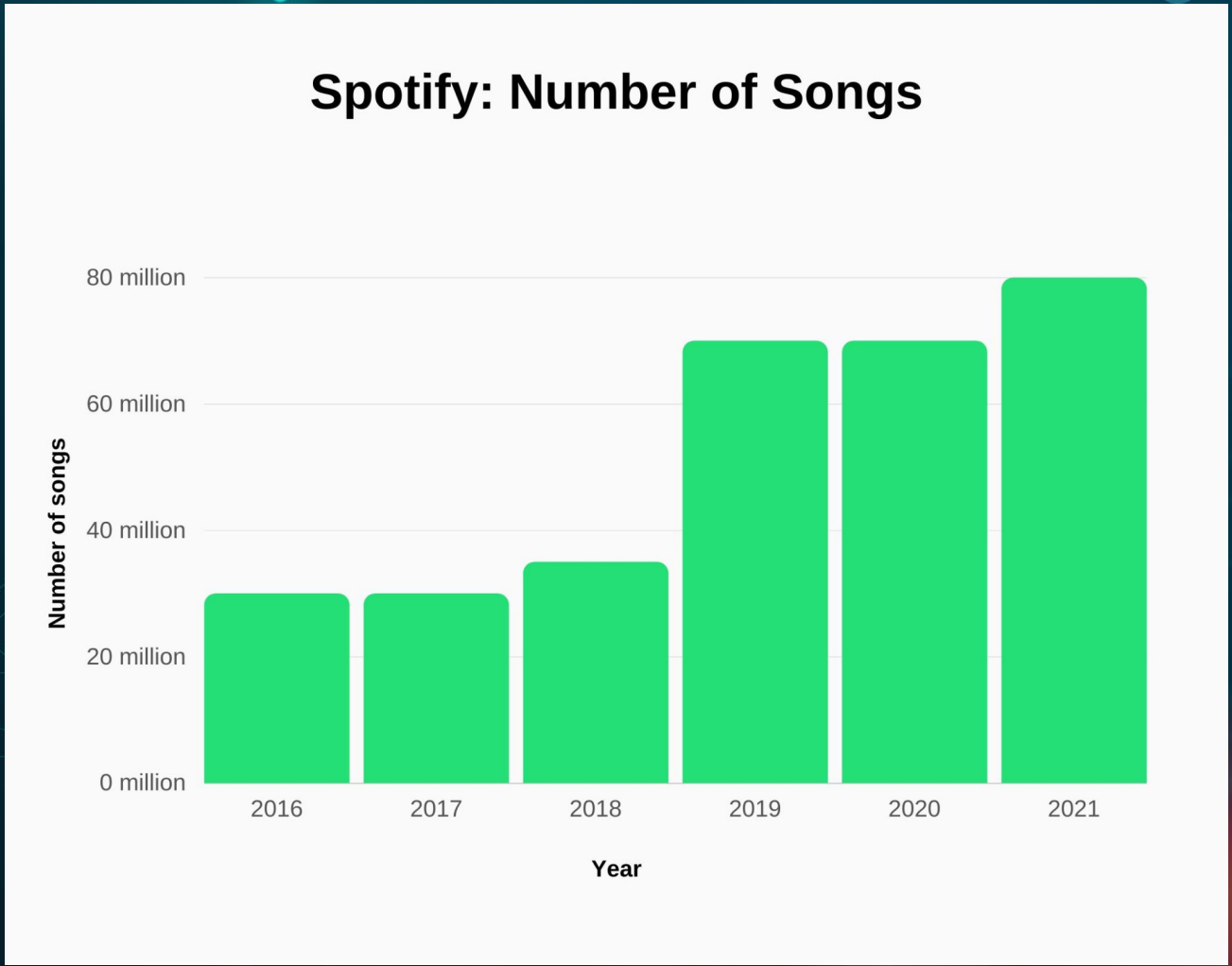
1. Audio Analysis and Features:

- Audio analysis (e.g., segments, bars, beats)
- Audio features (e.g., tempo, key, mode, loudness)
- Acoustic features (e.g., acousticness, instrumentalness)
- Danceability, energy, valence, and more

1. Categories and Genres:

- Browse categories
- Category playlists
- List of available genres

As reported in 2023, Spotify had over 100 million tracks on the platform.



Nuxt is an open source framework that makes web development intuitive and powerful.

Create performant and production-grade full-stack web apps and websites with confidence.

Features Of Nuxt.js Framework

- Nuxt.js, like Vue, is known among the developer community as one of the easiest frameworks to learn compared to other JavaScript-based frameworks.
- With Nuxt.js you can create fully scalable applications, thanks to the rendering options it offers.
- Thanks to its server-side rendering capabilities, it improves application performance.
- Enables rapid development.

RESULTS

User Input

A feature to Find Artists Based on their Popularity (less/more popular)

Marry go round in the beautiful field of sunflowers

Search

Use popularity

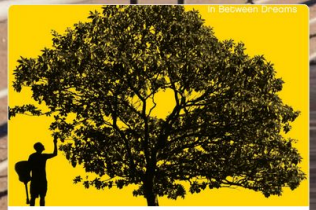
5

HOME ABOUT CONNECT

SENTIMENTAL RECOMMENDATIONS

Beat It - Single Version
Michael Jackson

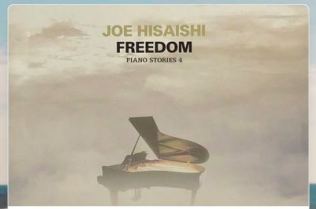
Listen | Copy Spotify Link



Banana Pancakes
Jack Johnson

Listen | Copy Spotify Link

SIMILAR RECOMMENDATIONS



Merry-Go-Round of Life - from 'Howl's Moving Castle'
Joe Hisaishi

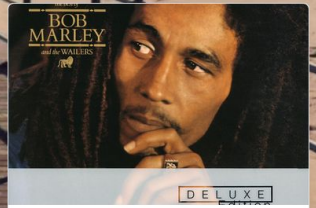
Listen | Copy Spotify Link

merry go round of life distorted remake
evan aloe xo

TARGETED POPULARITY

Waka Waka (This Time for Africa) [The Official 2010 FIFA World Cup (TM) Song] (feat. Freshlyground)
Shakira, Freshlyground

Listen | Copy Spotify Link



Could You Be Loved
Bob Marley & The Wailers

30°C Haze

ENG US

23:34
02-06-2024

Love birds flying in the winds of storm

Search

Use popularity

1

HOME ABOUT CONNECT

Listen | Copy Spotify Link

We R Who We R
Kesha

Listen | Copy Spotify Link

Lovebird Chirping, Pt. 4
Tom Green

Listen | Copy Spotify Link

All These Things That I've Done
The Killers

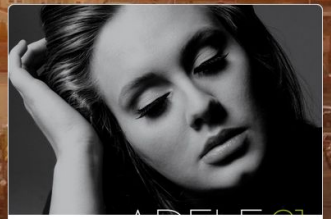
Listen | Copy Spotify Link

Viva La Vida
Coldplay

Listen | Copy Spotify Link

Lovebird Chirping, Pt. 5
Tom Green

Listen | Copy Spotify Link



Set Fire to the Rain
Adele

Listen | Copy Spotify Link

30°C Haze

ENG US

23:37 02-06-2024

http://localhost:3000/

HOME ABOUT CONNECT

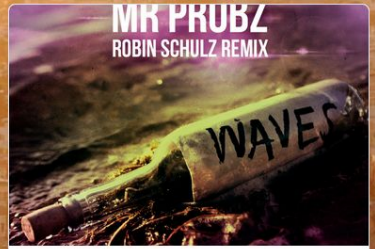
One Way or Another (Teenage Kicks)
One Direction
Listen | Copy Spotify Link

Lovebird Chirping, Pt. 1
Tom Green
Listen | Copy Spotify Link

Two Weeks
Grizzly Bear
Listen | Copy Spotify Link

Hard out Here
Lily Allen
Listen | Copy Spotify Link

Lovebird Chirping, Pt. 2
Tom Green
Listen | Copy Spotify Link



Waves - Robin Schulz Radio Edit
Mr. Probz, Robin Schulz
Listen | Copy Spotify Link

Addicted To You
Avicii
Listen | Copy Spotify Link

Lovebird Chirping, Pt. 3
Tom Green
Listen | Copy Spotify Link

Groove Is in the Heart
Deee-Lite
Listen | Copy Spotify Link

30°C Haze
ENG US
23:38 02-06-2024

CONCLUSION

We successfully developed a music recommendation application that uses sentiment analysis to personalize user experience. By integrating roBERTa for sentiment detection and using Spotify API for music recommendation we created a system that understands user sentiment and responds effectively. Using Nuxt.js, TypeScript , CSS Tailwind we endured a robust, responsive and user- friendly environment.

THANK YOU

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Ananya Sen
Konapalli Srinivas Arun
Sayan Mukherjee